

Write legibly showing all the steps VERY clearly. (Total points = 100).

Let $\Omega \neq \emptyset$ and $\mathcal{P}(\Omega)$ is the collection of all subsets of Ω , i.e the power set.

- (1) (a) Define what it means for $\mathcal{C} \subset \mathcal{P}(\Omega)$ to be an algebra.
 (b) Prove that $\mathcal{C}$ is an algebra iff $\mathcal{C}$ satisfies: $\Omega \in \mathcal{C}$ and $A, B \in \mathcal{C} \Rightarrow A \setminus B \in \mathcal{C}$
 (c) Let $\Omega = \{a, b, c, d\}$ and $\mathcal{A}_1 = \{\phi, \Omega, \{a, b\}, \{c, d\}\}$, $\mathcal{A}_2 = \{\phi, \Omega, \{a, c\}, \{b, d\}\}$.
 Check $\mathcal{A}_1$ and $\mathcal{A}_2$ are algebras. Is $\mathcal{A}_1 \cup \mathcal{A}_2$ an algebra? Explain.
 (d) Let $\mathcal{A}_n, n \geq 1$ be a sequence of *increasing* collection of algebras and $\mathcal{A} = \bigcup_{n=1}^{\infty} \mathcal{A}_n$.
 Prove that $\mathcal{A}$ is an algebra.

(5+15+15+15=50 points)

- (2) Let $(\Omega, \mathcal{F}, P)$ be a probability space and $X : \Omega \rightarrow \mathbb{R}$ be a random variable.

- (a) For $A, B \in \mathcal{F}$, if $P(A \Delta B) = 0$ then show $P(A) = P(B)$ [$A \Delta B = (A \setminus B) \cup (B \setminus A)$].
 (b) Let $B_i \in \mathcal{F}$ be such that $P(B_i) = 1$ for $i \geq 1$. Show $P(\cap B_i) = 1$.
 (c) Argue $\{\omega : \sin(X(\omega)) + \cos(X(\omega)) = 1\}$ is in $\mathcal{F}$.

(3×10 = 30 points)

- (3) Take $F(x) = x^3 \mathbf{I}_{(0,1)}(x) + \mathbf{I}_{[1,\infty)}(x), x \in \mathbb{R}$.

- (a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

[Write down the steps of the theorems carefully and state the results used in each step]

- (b) Why is such a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ unique?
 (c) If $A = (0.5, 2)$, then compute $\mu_F^*(A)$.

(10+5+5=20 points)

Write legibly showing all the steps VERY clearly. (Total points = 100).

- (1) (i) Let $P = m$ be the Lebesgue measure on $(\Omega = [0, 1], \mathcal{F} = \mathcal{B}([0, 1]))$ and $I_A(x)$ denote the indicator function of a set A . Consider for $n \geq 1$, $X_n(x) = nI_{[0, \frac{1}{n}]}(x)$ and $X(x) = 0$ for $x \in [0, 1]$.
- (a) Does $X_n \rightarrow X$ in probability (P)?
 - (b) Does $X_n \rightarrow X$ a.s (P)?
 - (c) Is the sequence of random variables $\{X_n\}$ uniformly integrable (UI)?
 - (d) For what values of $p \in (0, \infty)$ does $X_n \rightarrow X$ in L^p ?
- (ii) Let $X, X_n, n \geq 1$ be a sequence of random variables on *some* probability space $(\Omega, \mathcal{F}, P)$ such that for $n \geq 1$, $P(X_n = n) = \frac{1}{n} = 1 - P(X_n = 0)$ and $X = 0$ with probability 1. Without knowing the precise description of $(\Omega, \mathcal{F}, P)$ and the functional forms of $X, X_n, n \geq 1$, answer the following (with justification):
- (a) Can you say $X_n \rightarrow X$ in probability (P)?
 - (b) Can you say $X_n \rightarrow X$ a.s (P)?
 - (c) Can you say if the sequence of random variables $\{X_n\}$ is UI?
 - (d) Can you find out for what values of $p \in (0, \infty)$ does $X_n \rightarrow X$ in L^p ?

((8×8)=64 points)

- (2) Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and g, f, f_n are measurable real functions.
- (a) Show that, in general, $f_n \rightarrow f$ in L^p for some $0 < p < \infty$ implies that $f_n \rightarrow f$ in L^q , for any $0 < q < p$.
 - (b) Precisely **state** the *Dominated Convergence Theorem (DCT)* (Not the EDCT) and **prove** that the conclusion of DCT remains true even if " $f_n \rightarrow f$ a.e(μ)" is replaced by " $f_n \rightarrow f$ in measure" in the statement.
 - (c) Let $g, f_n \in L^1(\Omega, \mathcal{F}, \mu) \equiv L^1$ for $n \geq 1$. If $f_n \leq g$ for $n \geq 1$, then show that

$$\limsup_{n \rightarrow \infty} \int f_n d\mu \leq \int (\limsup_{n \rightarrow \infty} f_n) d\mu$$

(sometimes called the "reverse Fatou's lemma").

(10+(6+10)+10=36 points)